Ions Review Exercise

1. Which part of an atom can be gained or lost by the atom to form an ion?

2. If a neutral atom lost two electrons, what is the resulting charge?

3. If an atom had three more electrons than protons, what charge would it have?

4. Explain how an atom could develop a charge of +3

5. What are the names of the ions of the following elements?
    a. Sulphur
    b. Sodium
    c. Chlorine
    d. Magnesium
    e. Bromine

6. Name the following ions and state how many electrons have been gained or lost by each.
    a. Zn^{2+}
    b. S^{2-}
    c. O^{2-}
    d. Mg^{2+}
    e. Al^{3+}
    f. F^{-}

7. Explain how the periodic table can be used to predict the charge on some ions.

8. Write the ion symbols for
    a. Iron II and Iron III
    b. Lead II and Lead IV

9. What are the names of the following polyatomic ions?
    a. SO_4^{2-}
    b. OH^-
    c. PO_4^{3-}
    d. NO_3^-
    e. NH_4^+
    f. CO_3^{-2}

10. Write the symbol for each ion.
    a. Hydrogen
    b. Copper II
    c. Sulphide
    d. Iodide
    e. Lithium
    f. Phosphate
    g. Hydroxide
    h. Potassium
    i. Iron III
    j. Silver
    k. Magnesium
    l. Chloride
    m. Nitrate
    n. Carbonate
    o. Zinc
    p. Ammonium
    q. Oxide
    r. Sulphate
    s. Fluoride
    t. Calcium
    u. Aluminium
    v. Bromide
    w. Chromium
    x. Sodium
    y. Manganese
    z. Lead